

## TARAKAN, Indonesia

WMO: 965090

Lat: 3.3271N

Lon: 117.5700E

Elev: 6

StdP: 101.25

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 23.3       | 23.6 | 21.7                          | 16.3 | 26.6 | 22.2 | 16.8 | 26.4 | 6.4                   | 25.7 | 5.4 | 27.4 | 1.5                   | 310  | 0.326 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 5.7                    | 32.5            | 26.5 | 32.1 | 26.4 | 31.8 | 26.4 | 27.5                | 30.9 | 27.2 | 30.6 | 27.0 | 30.4 | 2.2                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.5                            | 22.1 | 29.5 | 26.2 | 21.7 | 29.1 | 26.1 | 21.5 | 29.0 | 87.3          | 31.3 | 86.1 | 31.0 | 85.1 | 30.7 | 30.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 5.4           | 4.4  | 3.9 | DB                         | 21.6 | 34.0               | 1.0 | 1.4                                                | 20.9 | 35.0       | 20.3 | 35.9       | 19.7 | 36.7       | 19.0 | 37.7 | (4) |
| (5)               |      |     | WB                         | 21.0 | 28.7               | 0.9 | 0.8                                                | 20.3 | 29.3       | 19.7 | 29.8       | 19.2 | 30.3       | 18.6 | 30.9 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 27.7        | 27.3  | 27.6  | 27.6  | 27.8  | 28.1  | 27.8  | 27.7  | 27.9  | 27.9  | 27.8  | 27.8  | 27.6  | (6)  |  |
|                                                                        | DBStd       | 0.96        | 0.97  | 0.90  | 0.99  | 0.87  | 0.84  | 0.99  | 0.97  | 1.03  | 0.96  | 0.96  | 0.82  | 0.93  | (7)  |  |
|                                                                        | HDD10.0     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD18.3     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (9)  |  |
|                                                                        | CDD10.0     | 6478        | 537   | 492   | 546   | 535   | 561   | 536   | 548   | 555   | 538   | 552   | 533   | 547   | (10) |  |
|                                                                        | CDD18.3     | 3437        | 279   | 258   | 288   | 285   | 302   | 286   | 289   | 297   | 288   | 294   | 283   | 288   | (11) |  |
|                                                                        | CDH23.3     | 36481       | 2738  | 2636  | 3023  | 3075  | 3359  | 3097  | 3069  | 3223  | 3124  | 3151  | 3008  | 2978  | (12) |  |
| (13)                                                                   | CDH26.7     | 11481       | 746   | 766   | 906   | 972   | 1126  | 1005  | 973   | 1081  | 1057  | 1041  | 926   | 884   | (13) |  |
| (14)                                                                   | Wind        | WSAvg       | 1.2   | 1.2   | 1.4   | 1.2   | 1.1   | 1.1   | 1.2   | 1.3   | 1.5   | 1.3   | 1.1   | 1.1   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 3320        | 260   | 240   | 285   | 320   | 313   | 260   | 265   | 276   | 265   | 343   | 327   | 350   | (15) |  |
|                                                                        | PrecMax     | 4722        | 517   | 429   | 521   | 517   | 754   | 482   | 573   | 601   | 542   | 667   | 592   | 588   | (16) |  |
|                                                                        | PrecMin     | 1754        | 57    | 46    | 34    | 10    | 49    | 47    | 4     | 25    | 31    | 138   | 12    | 67    | (17) |  |
|                                                                        | PrecStd     | 815         | 123   | 116   | 96    | 132   | 163   | 112   | 137   | 147   | 108   | 128   | 159   | 125   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 32.1  | 32.2  | 32.3  | 32.2  | 32.7  | 32.5  | 32.2  | 32.7  | 32.9  | 32.9  | 32.5  | 32.3  | (19) |  |
|                                                                        |             | MCWB        | 26.0  | 25.6  | 25.5  | 26.6  | 26.9  | 26.9  | 26.4  | 26.5  | 26.5  | 26.6  | 26.6  | 26.7  | (20) |  |
|                                                                        | 2%          | DB          | 31.2  | 31.4  | 31.6  | 31.7  | 32.0  | 31.9  | 31.6  | 32.0  | 32.0  | 32.0  | 31.7  | 31.6  | (21) |  |
|                                                                        |             | MCWB        | 26.0  | 25.6  | 25.7  | 26.5  | 26.7  | 26.7  | 26.3  | 26.3  | 26.3  | 26.4  | 26.6  | 26.5  | (22) |  |
|                                                                        | 5%          | DB          | 30.7  | 30.9  | 31.0  | 31.2  | 31.4  | 31.2  | 31.1  | 31.3  | 31.4  | 31.3  | 31.1  | 31.0  | (23) |  |
|                                                                        |             | MCWB        | 25.9  | 25.6  | 25.9  | 26.4  | 26.7  | 26.6  | 26.3  | 26.1  | 26.2  | 26.4  | 26.5  | 26.4  | (24) |  |
|                                                                        | 10%         | DB          | 30.1  | 30.2  | 30.4  | 30.6  | 30.8  | 30.6  | 30.5  | 30.7  | 30.8  | 30.7  | 30.5  | 30.4  | (25) |  |
|                                                                        |             | MCWB        | 25.7  | 25.6  | 25.9  | 26.3  | 26.6  | 26.4  | 26.1  | 26.0  | 26.2  | 26.3  | 26.3  | 26.2  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 27.0  | 26.8  | 27.2  | 27.5  | 27.7  | 27.7  | 27.2  | 27.2  | 27.6  | 27.6  | 27.5  | 27.3  | (27) |  |
|                                                                        |             | MCDB        | 30.6  | 30.1  | 30.4  | 30.9  | 31.0  | 31.0  | 30.5  | 30.9  | 31.2  | 31.0  | 30.9  | 30.8  | (28) |  |
|                                                                        | 2%          | WB          | 26.6  | 26.4  | 26.7  | 27.1  | 27.3  | 27.2  | 26.8  | 26.8  | 27.1  | 27.1  | 27.1  | 27.0  | (29) |  |
|                                                                        |             | MCDB        | 29.9  | 29.7  | 29.9  | 30.5  | 30.5  | 30.5  | 30.1  | 30.2  | 30.6  | 30.6  | 30.5  | 30.4  | (30) |  |
|                                                                        | 5%          | WB          | 26.2  | 26.1  | 26.4  | 26.7  | 27.0  | 27.0  | 26.6  | 26.6  | 26.7  | 26.7  | 26.7  | 26.7  | (31) |  |
|                                                                        |             | MCDB        | 29.4  | 29.3  | 29.5  | 29.9  | 30.2  | 30.1  | 29.8  | 30.0  | 30.0  | 30.1  | 30.0  | 29.9  | (32) |  |
|                                                                        | 10%         | WB          | 26.0  | 25.9  | 26.1  | 26.5  | 26.7  | 26.6  | 26.3  | 26.3  | 26.4  | 26.5  | 26.5  | 26.4  | (33) |  |
|                                                                        |             | MCDB        | 29.0  | 28.9  | 29.1  | 29.5  | 29.7  | 29.6  | 29.4  | 29.5  | 29.7  | 29.7  | 29.6  | 29.4  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.5   | 5.5   | 5.4   | 5.5   | 5.7   | 5.7   | 5.9   | 6.0   | 6.2   | 6.1   | 5.6   | 5.6   | (35) |  |
|                                                                        |             | MCDBR       | 6.5   | 6.3   | 6.4   | 6.3   | 6.4   | 6.4   | 6.5   | 6.5   | 6.9   | 6.9   | 6.5   | 6.4   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 2.2   | 2.1   | 2.2   | 2.4   | 2.4   | 2.5   | 2.6   | 2.5   | 2.7   | 2.7   | 2.5   | 2.5   | (37) |  |
|                                                                        |             | MCWBR       | 5.9   | 5.7   | 5.8   | 5.8   | 5.9   | 6.1   | 6.2   | 6.3   | 6.5   | 6.5   | 6.0   | 6.1   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.398 | 0.407 | 0.408 | 0.410 | 0.399 | 0.400 | 0.404 | 0.454 | 0.480 | 0.505 | 0.415 | 0.397 | (40) |  |
|                                                                        |             | taud        | 2.553 | 2.504 | 2.504 | 2.492 | 2.542 | 2.537 | 2.518 | 2.350 | 2.270 | 2.199 | 2.482 | 2.557 | (41) |  |
|                                                                        | Ebn at Noon |             | 927   | 922   | 916   | 897   | 886   | 873   | 873   | 846   | 839   | 823   | 903   | 922   | (42) |  |
|                                                                        |             | Edh at Noon | 105   | 112   | 113   | 111   | 103   | 101   | 104   | 127   | 140   | 150   | 112   | 103   | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 4.48  | 5.12  | 5.03  | 5.24  | 5.12  | 4.81  | 4.93  | 5.26  | 5.24  | 4.96  | 4.76  | 4.60  | (44) |  |
|                                                                        | RadStd      |             | 0.33  | 0.45  | 0.46  | 0.39  | 0.27  | 0.22  | 0.36  | 0.26  | 0.32  | 0.34  | 0.28  | 0.31  | (45) |  |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (2 neighbors) | +0.17 | N/A     | N/A    | +0.36   | +0.47 | +0.42 | N/A         | N/A     | +121    | +121    |

Nomenclature: See separate page